TANYA VERMA

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EDUCATION

Thapar Institute of Engineering and Technology **Aug 2023 – May 2027**
B.Tech. in Computer Engineering | CGPA: 7.94/10 Patiala, Punjab
Class XII (CBSE): 91.6%

EXPERIENCE

Software Development Engineer Intern – AWFIS **Jun 2025 – Jul 2025**
Chatbot Backend and Client-Server Platform
- Developed Python and REST-based chatbot backend workflows handling 120+ daily customer queries, improving response efficiency by 30% through request-processing optimizations.
- Implemented asynchronous request handling and concurrent workloads, reducing API latency by 25% and improving backend responsiveness.
- Built and tested RESTful APIs, database persistence, and client-server workflows supporting 50+ concurrent users.

AI Intern – TechnoVentra **Jun 2026 – Jul 2026**
Document Intelligence and Data Processing Automation
- Developed Python-based data-processing pipelines to extract, transform, validate, and structure unstructured PDF content for downstream AI/ML applications.
- Automated information extraction and data-validation workflows, reducing manual extraction effort by 80%.
- Prepared structured data for downstream knowledge-graph construction and LLM-based retrieval applications.

PROJECTS

High-Performance DPI Engine – Network Processing [GitHub] **Jan 2026 – Mar 2026**
C++17, Data Structures, OOP, Multithreading, TCP/IP, TCP/UDP, TLS, DNS, QUIC, CMake
- Designed a multithreaded producer-consumer architecture using bounded queues, concurrent packet processing, and deterministic flow-to-worker assignment.
- Implemented packet parsing, five-tuple flow hashing, TCP connection tracking, and protocol handling for TCP/UDP, TLS, DNS, and QUIC.
- Achieved 1M+ packets/sec at 1+ GB/s throughput, with 99%+ test-case correctness and 90%+ macro-F1 across six traffic categories.

Multi-Agent AutoML Data Analyst – Autonomous AI Platform [GitHub] **Aug 2026 – Sep 2026**
Python, FastAPI, Scikit-learn, Gemini, MCP, A2A, Multi-Agent Systems, Streamlit, Pandas, NumPy, nbformat
- Built an autonomous multi-agent AI application orchestrating 5 specialized roles—Profiler, EDA, AutoML, Verifier, and Notebook Synthesizer—with Gemini as the LLM backbone, Agent-to-Agent (A2A) communication, and persistent agent memory.
- Automated end-to-end ML workflows across 5 models (3 regression, 2 classification), including preprocessing, feature engineering, outlier handling, feature selection, cross-validation, hyperparameter tuning, and evaluation.
- Integrated MCP-based tools for dataset analysis, visualization, feature engineering, file operations, and Jupyter notebook generation, powering an 8-page Streamlit application with AI-generated explanations and recommendations.

Cloud-Native Full-Stack Application – AWS [GitHub] **Jul 2026 – Aug 2026**
JavaScript, React.js, Node.js, Express.js, PostgreSQL, AWS, Docker, Terraform, CI/CD
- Designed, developed, and deployed a containerized full-stack application using React.js, Node.js/Express, and PostgreSQL on AWS ECS Fargate behind an Application Load Balancer.
- Provisioned cloud infrastructure including VPC, subnets, security groups, RDS, ECS, ECR, IAM, Secrets Manager, and CloudWatch using Terraform.
- Implemented CI/CD with GitHub Actions and AWS OIDC to run automated checks, publish Docker images, update ECS task definitions, and validate deployments.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, JavaScript, SQL
Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Multithreading
AI/ML: Machine Learning, Generative AI, LLMs, NLP, AutoML, Scikit-learn, XGBoost, CatBoost, SHAP, Pandas, NumPy
RAG & Agents: RAG, AI Agents, LangChain, LangGraph, Gemini, MCP, A2A, Prompt Engineering, Embeddings, Vector Databases, ChromaDB
Backend & APIs: FastAPI, REST APIs, API Integration, Backend Development, Client-Server Architecture, Automation Workflows
Cloud & DevOps: AWS, ECS Fargate, ECR, RDS, VPC, IAM, Terraform, Docker, GitHub Actions, CI/CD, CloudWatch, GCP, Google Cloud Run, Linux, Git, GitHub

LEADERSHIP EXPERIENCE

Marketing Coordinator – Microsoft Learn Student Chapter, TIET **Aug 2024 – May 2025**
- Led technology-focused outreach campaigns across LinkedIn and Discord, reaching 1,000+ students.
- Coordinated cross-functional teams to deliver technical content, promotional campaigns, and student-facing events.

ACHIEVEMENTS

Smart India Hackathon: Shortlisted among institute-level teams from 500+ participating teams for an AI-based Rockfall Prediction and Early Warning System for Open-Pit Mines.